Tarea 4

Ponderación: 12% de la nota final del curso Formato: Desarrollar esta tarea en un`Rmarkdown`.

Instrucciones:

Usa el siguiente cógigo para cargar la base de datos sobre Covid-19 usados en clase:

```
library("tidyverse")
library("wesanderson")
library("cowplot")
library("lubridate")
library("viridis")
library("ggdark")

path <- url("https://raw.githubusercontent.com/mebucca/dar_soc4001/master/slides/class_12/covid_data.cs
# leer archivo csv
covid_data <- read_delim(path, delim=";")
```

Referencia: Hasell, J., Mathieu, E., Beltekian, D. et al. A cross-country database of COVID-19 testing. Sci Data 7, 345 (2020). Link

Los datos deben verse así:

```
## Rows: 56,748
## Columns: 50
## $ iso_code                          <chr> "AFG", "AFG", "AFG", "AFG", "AFG", ~
## $ continent                         <chr> "Asia", "Asia", "Asia", "Asia", "As~
## $ location                          <chr> "Afghanistan", "Afghanistan", "Afgh~
## $ date                              <date> 2019-12-31, 2020-01-01, 2020-01-02~
## $ total_cases                       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_cases                         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ new_cases_smoothed                <dbl> NA, NA, NA, NA, NA, NA, 0, 0, 0, 0,~
## $ total_deaths                      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_deaths                        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ new_deaths_smoothed               <dbl> NA, NA, NA, NA, NA, NA, 0, 0, 0, 0,~
## $ total_cases_per_million           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_cases_per_million             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ new_cases_smoothed_per_million    <dbl> NA, NA, NA, NA, NA, NA, 0, 0, 0, 0,~
## $ total_deaths_per_million          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_deaths_per_million            <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ new_deaths_smoothed_per_million   <dbl> NA, NA, NA, NA, NA, NA, 0, 0, 0, 0,~
## $ reproduction_rate                 <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ icu_patients                      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ icu_patients_per_million          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ hosp_patients                     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ hosp_patients_per_million         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```

```
## $ weekly_icu_admissions              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ weekly_icu_admissions_per_million  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ weekly_hosp_admissions             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ weekly_hosp_admissions_per_million <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ total_tests                        <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_tests                          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ total_tests_per_thousand           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_tests_per_thousand             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_tests_smoothed                 <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ new_tests_smoothed_per_thousand    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ tests_per_case                     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ positive_rate                      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ tests_units                        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ stringency_index                   <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ population                         <dbl> 38928341, 38928341, 38928341, 38928~
## $ population_density                 <dbl> 54.422, 54.422, 54.422, 54.422, 54.~
## $ median_age                         <dbl> 18.6, 18.6, 18.6, 18.6, 18.6, 18.6,~
## $ aged_65_older                      <dbl> 2.581, 2.581, 2.581, 2.581, 2.581, ~
## $ aged_70_older                      <dbl> 1.337, 1.337, 1.337, 1.337, 1.337, ~
## $ gdp_per_capita                     <dbl> 1803.987, 1803.987, 1803.987, 1803.~
## $ extreme_poverty                    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ cardiovasc_death_rate              <dbl> 597.029, 597.029, 597.029, 597.029,~
## $ diabetes_prevalence                <dbl> 9.59, 9.59, 9.59, 9.59, 9.59, 9.59,~
## $ female_smokers                     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ male_smokers                       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ handwashing_facilities             <dbl> 37.746, 37.746, 37.746, 37.746, 37.~
## $ hospital_beds_per_thousand         <dbl> 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, ~
## $ life_expectancy                    <dbl> 64.83, 64.83, 64.83, 64.83, 64.83, ~
## $ human_development_index            <dbl> 498, 498, 498, 498, 498, 498, 498, ~
```

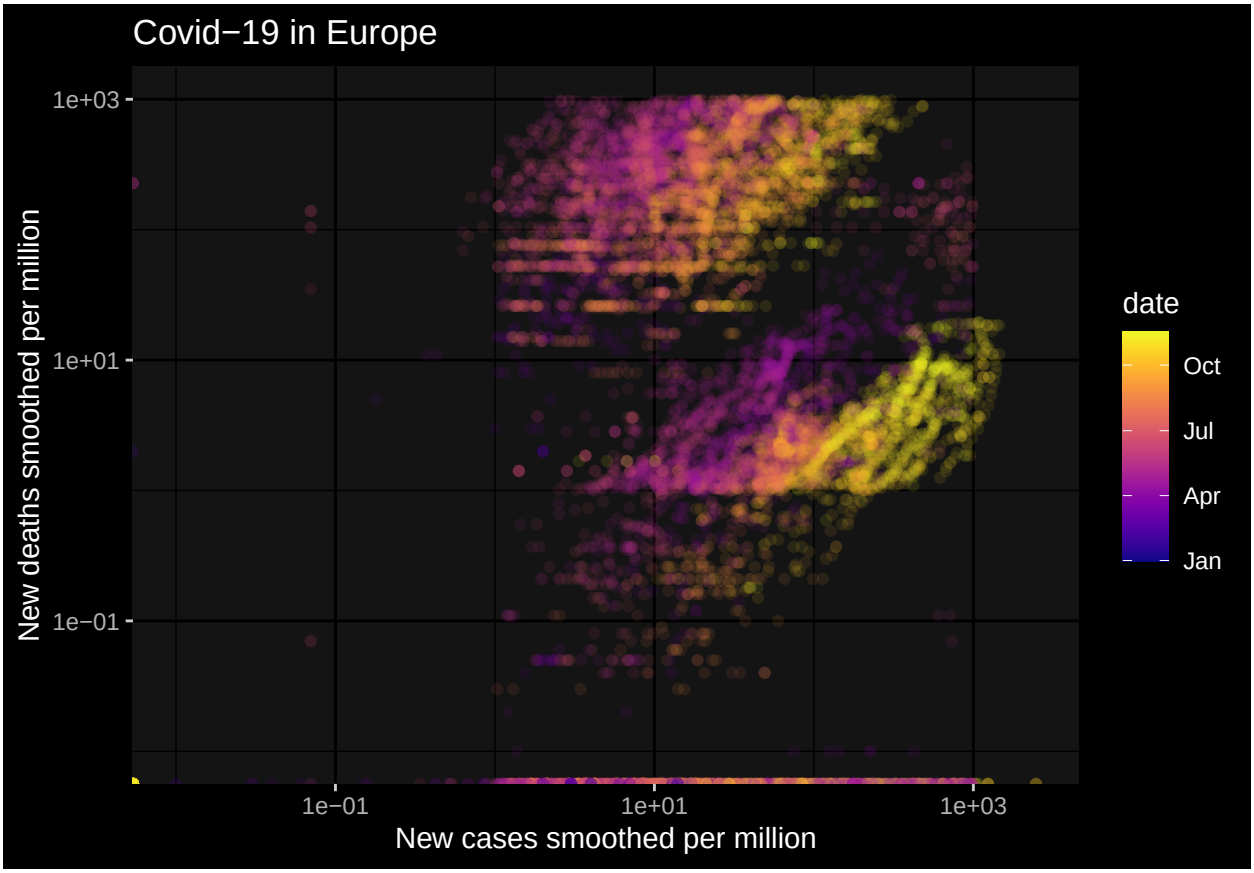
Ejercicio

Usando `Rmarkdown`, produce un reporte lo más parecido posible al documento que estás leyendo.

La figura a continuación usa un `theme` y una paleta de colores de tu preferencia. **Pista**: mi paleta de colores está definida por la siguiente linea de código: `scale_color_viridis(trans = "date", option = "plasma")` y mi theme es `dark_theme_gray()`, del paquete `library("ggdark")`. Ambos ejes están graficados en escala log10.

```r
plot <- covid_data  %>% filter(continent=="Europe") %>%
  ggplot(aes(x=new_cases_smoothed_per_million, y=new_deaths_smoothed_per_million, colour=date)) +
  geom_point(alpha=0.1) +
  scale_x_log10() +  scale_y_log10() +
  dark_theme_gray() +
  scale_color_viridis(trans = "date", option = "plasma") +
  labs(x="New cases smoothed per million", y="New deaths smoothed per million", title="Covid-19 in Europ

print(plot)
```



En el periodo estudiado el promedio de casos nuevos por millón de habitantes en Europa fue de 96.67, mientras que el promedio de muertes nuevas por millón de habitantes fue 86.